

Elastic Compute Cloud (Ec2)

Amazon ec2 provides scalable computing Capacity in the AWS Cloud

★ ★ Virtual machine , instance , virtual server => all are same

you can create Amazon ec2 to launch as many or few virtual server as you need
configure security and networking and manage storage

Amazon ec2 enable you to scale up or scale down the instance

Amazon ec2 is having 2 storage option i.e EBS , and instance store(fast)

Pre –Configured templates are available is know as Amazon machine Image

By default when we create an ec2 account with the amazon your account is limited to maximum of 20 instance per ec2 region with 2 default high i/o instance

Type of Ec2 Instance

- 1 . General purpose instance - > balance memory and cpu
- 2 . Compute Optimized - > more cpu than ram
- 3 . Memory Optimized - > More Ram
- 4 . Storage Optimized - > low latency
- 5 . Accelerated Computing / GPU - > Graphic Optimized
- 6 . High Memory - > High Ram nitro System

General Purpose Instance

General purpose instance provide balance of compute memory and networking resource and can be used for a variety of workloads

3 series come under the general purpose instance

size -> nano , small , medium , large

A series -> size (larger , medium)

T series -> T2 -> this is free tier

T3

T3a

M series -> M4 : only large size is available in m series 🙋

M5

M5a , m5ad , m5d

A series

A1 instance are ideally suited for **scale-out workloads** that are supported by arm Ecosystem.

These instance are well suited for the following Application

- 1 . **Web-server**
- 2 . Caching fleets
- 3 . **Containerised micro services**
- 4 . Distributed data store
- 5 . Application that required arm instruction set

M4 instance

The new m4 instance features a custom intel xeon e5-2676 v3 has well processor
Optimized specifically for ec2

VCPU -> 2 to 40 max

Ram -> 8gb to 160gb (max)

Instance storage only EBS

These instance are well suited for the following Application

T Series

T2 t3 . And t3a INSTANCE

These instance provide a **baseline level** of **cpu performance with the ability to burst to a higher level** when required by your workload

An **unlimited instance can sustain high cpu performance** for any period of time
Whenever required

USED FOR :)

1. Website and web app
- 2 . Code Repositories
- 3 . Development , build , test
- 4 . Microservices

Vcpu -> 2 to 8

Ram -> **0.5 to 32 GB**

2 . Compute Optimized Instance

only c series is available

Compute optimized instance are ideal for compute –bound application that benefit from high performance processor

Three Types are Available

C4 , C5 , C5n

C4 ☺ C4 Instance are optimized for compute intensive workload and deliver very cost effective high performance at a low price compute ratio

vcpu 2 to 36

Ram -> 3.75 to 60 GB

Storage -> EBS only

Network -> 10gb

uses ☺ web server , Batch processing , mmo gaming , video encoding

3 . Memory Optimized Instance

3 series is available in this instance

Memory optimized instance are designed to deliver fast performance for workloads that process large data sets in memory

R series

High performance Relational (mysql) , nosql (mongodb , Cassandra) database

Distributed web scale cache store that provide type data used in financial services Hadoop

vcpu 2 to 96

Ram 16 gb to 768gb

instance storage only ☺ EBS only and Nvme SSD

X series x1 , x1e

Well suited for high performance database , memory intensive enterprise application relational database , workload sap Hana

Electric Design Automation

Vcpu 4 to 128

Ram 122 to 3904 GB

Instance storage -> ssd

Z1d instance

High Frequency Z1d delivers a sustained all core frequency of upto 4.0ghz the fastest of any cloud instance .

Aws nitro system xeon processor up to 1.8tb of instance storage

Vcpu 2 to 48

Ram 16 to 384gb

Storage nvm SSD

useCase -> Electronic Design

Automation , and Certain database and certain database workloads with high per-core licensing Cost

4 . Storage Optimized Instance

3 three series is present in this instance (IDH SERIES)

Storage Optimized instance are designed for workloads that require high sequential read and write access to very large data sets on local storage

They are Optimized to deliver tens for thousand of low latency Random I/O Operations per second (iops) to application

D2 instance

Well suited for the folloing

- 1 . massive parallel processing (MMP) data warehouse
- 2 . Map Reduce and Hadoop distributed computing
- 3 . Log or data Processing app

Vcpu 4 to 36

Ram 30.5 to 244 GB

Storage ssd

H1 instance

The family features up to 16tb of HDD based local storage high disk throughput and balance of compute & memory

Well suited for App Requiring Sequential access to large amount of data on direct attached instance storage

Application that Requires high throughput access to large Quantities of data
Quantities of data

vcpu 8 to 64

Ram 32 to 256 Gb

Storage HDD

I3 and I3en instance

Well suited for :

- 1 .High Frequency online transaction processing system (OLTP)
- 2 .Relational database
- 3 .No-sql database
- 4 . Distributed file system
- 5 . Data warehousing Application

vcpu 2 to 96

Ram 16 to 768gb

local storage Nvme SSD

Networking performance 25gb to 100gbps

Read -> 16gb/s (i3) , 8gb/s (I3en)

5 . Accelerated Computing Instance

3 three series is present in this instance (PGF SERIES)

Accelerated computing instance families use hardware

Accelerated or co-processor to perform some function such as floating point number calculation , graphics , processing , or data , pattern matching more efficiently than is possible in software running on cpu.

F1 Instance

F1 instance offer customizable hardware acceleration with

Field Programmable Gate array (fpga)

Each Fpga contains 2.5millions logic elements & 8600 dsp engine

Design to accelerate computational intensive algorithms such as data flow or highly parallel operations

F1 provides local nvme ssd storage

vcpu -> 8 to 64

fpga -> 1 to 8

Ram -> 122 to 976

Storage -> nvme ssd

Used in geonomics research financial analytics real time video processing and big data search

P2 and P3 Instance

it is use Nvidia tesla GPUs

Provides high Bandwidth networking

up to 32 gb of memory per gpu which , make them ideal for deep

Learning and computational fluid dynamic

p2
vcpu - > 4 to 64

gpu -> 1 to 16

Ram -> 61 to 732gb

gpu Ram -> 12 – 192 Gb

p3
vcpu -> 8 to 96

Gpu -> 1 to 8

Ram ->61 to 768Gb

Storage -> ssd and ebs

G2 and G3 Instance

Optimized for Graphics intensive application
well suited for app like 3d visualisation

G3 instance use NVidia tesla

m60 Gpu , and provide a cost effective high performance
platform for graphics Application

Vcpu -> 4 to 64

Gpu -> 1 to 4

Ram -> 30.5 to 488gb

Gpu -> 8 to 32 gb

Network performance 25 gbps

6 . High Memory Instance

important key points

High Memory Instance are bare metal instance(directly install on hardware) and do not run on a

Hypervisor

Only available under dedicated host purchasing category
(for 3 year term)

O.S directly on hardware

High memory instance are purpose built to run large in memory database including production development of sap Hana in the cloud

Features

Latest generation intel Xeon Pentium 8176m Processor

6 , 9 , 12 Tb of instance memory the largest of any ec2 instance

Powered by the Aws nitro system a combination of dedicated hardware & lightweight hypervisor

Based metal performance with direct access to host hardware
EBS optimized by default at no additional cost

Model number , u-6TB1-metal , U 1tb1.metal & U12tb1.metal

Network performance 25gb , dedicated EBS Bandwidth 14gbs
Each Instance offers 448 logical processor

7 . Previous Generation Instance

important key points

T1 , M1 , C1 , CC2 , M2 , CRI , CG1 , I2 , HIS ,M3 , C3 , and R3

Instance Purchasing Options

1 . On Demand

2 . On Dedicated instance

3 . Dedicated Host

4 . Spot Instance

5 . Schedule Instance

6 . Reserved Instance -> Standard Ri , Convertible Ri , Schedule Ri

There are 4 ways to pay Amazon Ec2 instance i.e On-demand
Reserved Instance and spot Instance

1 . On Demand Purchasing Options